

A From-Scratch Reproducibility and Ablation Study of Lite Fourier Neural Operator

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Abstract

We present a from-scratch reproducibility and ablation study of *LiteFNO* (Ahn et al., 2025), a lightweight Fourier Neural Operator proposed for time-dependent PDEs that claims 99.9% parameter reduction and $\sim 44\%$ VRMSE improvement over FNO variants. Because no public implementation was available at the time of this study, we independently reimplement two architectures from the paper description: (A) a CP-factorized spectral LiteFNO via the **neuraloperator** library, and (B) a low-rank CNN ablation baseline that replaces Fourier layers with 1×1 and 3×3 convolutional bottlenecks. Our central question is: *does LiteFNO’s Fourier machinery earn its architectural complexity, or does a plain low-rank CNN reach the same lightweight-efficiency frontier?* We evaluate on the Gray-Scott reaction-diffusion dataset from The Well (Ohana et al., 2024) under a controlled head-to-head protocol, reporting test VRMSE of 0.0126 (spectral LiteFNO, 179,666 parameters, 169.3 MFLOPs/forward) vs. 0.0151 (low-rank CNN, 107,842 parameters, 218.6 MFLOPs/forward). The two models are closely matched on accuracy, but occupy different points on the efficiency frontier: LiteFNO is more FLOP-efficient; the CNN is more parameter-efficient and $3\times$ faster on GPU due to lower FFT overhead. Extensions covering quantization (both lossless to int8, with LiteFNO degrading more gracefully at int4), autoregressive rollout (LiteFNO accumulates error more slowly at the 6:12 and 13:30 windows), noise robustness, spectral sensitivity, and compute benchmarks reveal where the Fourier inductive bias provides meaningful practical value. We also document the implementation ambiguities encountered during reimplementations as actionable lessons for future practitioners. All code is available at <https://anonymous.4open.science/r/litefno-repro-C05C/>.

1 Introduction and Motivation

Neural operators have emerged as a powerful paradigm for learning mappings between function spaces, enabling data-driven surrogates for partial differential equations (PDEs) that are orders of magnitude cheaper than numerical solvers (Kovachki et al., 2023; Li et al., 2021). The Fourier Neural Operator (FNO) (Li et al., 2021) parameterizes integral kernels in the Fourier domain, providing a global receptive field and theoretical resolution-invariance. These properties make FNO-style models appealing for scientific computing, but their parameter counts can be prohibitive in low-resource settings.

Ahn et al. (2025) address this with LiteFNO, which combines spectral convolution with CP low-rank factorization of the spectral weights and a “transduction” mechanism for spatio-temporal extension, reporting 99.9% parameter reduction and $\sim 44\%$ VRMSE improvement over competing FNO variants on eight datasets from The Well (Ohana et al., 2024). These efficiency claims are directly relevant to practitioners in low-resource settings such as climate-adaptation decision support, where both model footprint and inference cost matter.

In attempting to reproduce LiteFNO, we found that its architectural description, while detailed, requires non-trivial implementation choices: the CP factorization interface in **neuraloperator** exposes fractional rather than integer rank, training protocol details are underspecified, and the transduction initialization is described only in an appendix. We document these ambiguities systematically. We also ask a secondary architectural

question: is LiteFNO’s Fourier machinery actually necessary for competitive lightweight performance, or does a plain convolutional bottleneck suffice? We answer this with a two-arm controlled comparison.

Our study is structured as MLRC Category 2 (generalization and ablation). We do not claim to reproduce the paper’s exact numbers, which require a different protocol (500 epochs, relative- L^2 loss, transduction, 64×64 resolution); instead, we answer the relative architectural question under a matched head-to-head protocol and document what the spectral inductive bias actually buys under a controlled comparison.

Contributions.

1. A from-scratch implementation of spectral LiteFNO using the `neuraloperator` library, fully documented and released (§4), together with a systematic account of implementation ambiguities in the original paper.
2. A low-rank CNN ablation baseline matched in parameter budget, isolating the contribution of the Fourier layers (§4).
3. A controlled two-arm head-to-head comparison on Gray-Scott, showing that LiteFNO and the CNN occupy different but nearby points on the accuracy–efficiency Pareto frontier (§6).
4. Extensions probing quantization, noise robustness, autoregressive rollout, spectral sensitivity, and compute efficiency, identifying where the Fourier inductive bias provides practical value (§7).

2 Related Work

Fourier Neural Operators. Li et al. (2021) introduced FNO; subsequent work has extended it to irregular geometries, multiscale problems, and coupled systems (Kovachki et al., 2023). Tensorized variants (Kossaifi et al., 2023) apply CP, Tucker, or tensor-train decompositions to the spectral weight tensors, the strategy LiteFNO adopts. Our spectral reimplement uses the `neuraloperator` library’s `TFNO` class, which exposes these factorizations.

Lightweight surrogates. Bottleneck CNNs, separable convolutions, and knowledge distillation have all been applied to reduce surrogate-model footprint. Our CNN ablation can be seen as a well-tuned bottleneck CNN baseline; its competitive performance suggests Fourier inductive biases may be less critical than parameter count at the spatial scales studied.

Scientific ML benchmarks. The Well (Ohana et al., 2024) provides standardized PDE datasets enabling fair cross-model comparison. We use its Gray-Scott (GS) split as our primary evaluation domain.

From-scratch reproducibility. Reproducing a paper’s results by reimplementing from the description, rather than running released code, tests whether the paper’s description is sufficient for an independent practitioner and surfaces ambiguities a code release would hide (Pineau et al., 2021). Our study provides this for LiteFNO.

3 Background

3.1 Fourier Neural Operator

Let $u : \mathcal{D} \rightarrow \mathbb{R}^{d_u}$ denote a discretized field on a spatial domain $\mathcal{D} \subset \mathbb{R}^d$. The FNO (Li et al., 2021) is a stack of L Fourier layers:

$$v^{(\ell+1)}(\mathbf{x}) = \sigma\left(\mathcal{F}^{-1}\left[R^{(\ell)} \cdot \mathcal{F}[v^{(\ell)}]\right](\mathbf{x}) + W^{(\ell)}v^{(\ell)}(\mathbf{x})\right), \quad (1)$$

where $\mathcal{F}$ is the truncated FFT, $R^{(\ell)} \in \mathbb{C}^{k_{\max} \times d_v \times d_v}$ is the learnable spectral kernel, $W^{(\ell)}$ is a pointwise skip connection, and σ is a nonlinearity.

3.2 CP Low-Rank Factorization of Spectral Weights

The spectral weight tensor $R \in \mathbb{C}^{k_1 \times \dots \times k_d \times d_v \times d_v}$ is expensive in high dimensions. Ahn et al. (2025) factorize it in the CANDECOMP/PARAFAC (CP) sense:

$$R \approx \sum_{r=1}^{\rho} \mathbf{a}_r^{(1)} \otimes \dots \otimes \mathbf{a}_r^{(d)} \otimes \mathbf{b}_r \otimes \mathbf{c}_r, \quad (2)$$

where ρ is the CP rank and $\otimes$ denotes the outer product. This reduces spectral parameter count from $O(mnlk^d)$ to $O((m+n+kd)l\rho)$, enabling the claimed 99.9% reduction relative to full FNO.

3.3 Transduction for Spatio-Temporal Extension

LiteFNO extends the spatial model to spatio-temporal fields by adding a temporal factor to the CP decomposition and fine-tuning for 100 epochs with the spatial weights frozen. Grid embeddings are extended from 2D to 3D, with new lifting weights initialized to zero and new spectral factors initialized via conjugate symmetry to avoid disrupting the pretrained spatial weights. We do not implement transduction in our matched-protocol experiments (see §9); it remains a direction for future work.

3.4 The Low-Rank CNN Ablation

Our ablation arm replaces each FNO Fourier layer with a bottleneck block:

$$v^{(\ell+1)} = \text{GELU}\left(\mathbf{W}_{\text{exp}} \text{GELU}\left(\mathbf{W}_{3 \times 3} \text{GELU}\left(\mathbf{W}_{\text{red}} v^{(\ell)}\right)\right)\right) + v^{(\ell)}, \quad (3)$$

where $\mathbf{W}_{\text{red}}$ and $\mathbf{W}_{\text{exp}}$ are 1×1 convolutions and $\mathbf{W}_{3 \times 3}$ is a 3×3 convolution. This introduces a convolutional low-rank bottleneck in feature space but performs no Fourier transform, providing a clean ablation of the spectral component.

4 Method: Two-Arm Study Design

We define two experimental arms, trained and evaluated under the identical protocol described in §5.

Arm A: Spectral LiteFNO (from paper). We implement LiteFNO as described in Ahn et al. (2025) using the `neuraloperator` library (Kossaifi et al., 2023). The model uses $L = 8$ TFNO2d layers with CP factorization at fractional rank $\rho = 0.02$, $k_{\text{max}} = 16$ Fourier modes per dimension, and $d_v = 64$ hidden channels, yielding 179,666 parameters.

One implementation gap required resolution: `neuraloperator` exposes CP rank as a fraction of the full tensor rank, whereas the paper reports integer ranks $\{32, 48\}$. Our choice of $\rho = 0.02$ matches the paper’s approximate parameter budget as closely as practicable, but the exact correspondence cannot be guaranteed without knowing the full tensor dimensions, which depend on architectural details not fully specified in the paper. AMP was *disabled* for this arm: `neuraloperator`’s complex-weight layers are incompatible with PyTorch’s `GradScaler`, causing NaN losses under fp16; Arm A therefore trained in fp32 (see §9).

Arm B: Low-rank CNN ablation. We implement the bottleneck CNN described in §3.4: 8 residual blocks each consisting of a 1×1 channel-reduction convolution, 3×3 spatial convolution, and 1×1 channel-expansion convolution with GELU activations. This arm has 107,842 parameters and serves as the “no Fourier” control, testing whether spectral inductive bias is necessary for competitive lightweight performance.

Why this design answers the research question. Arm A tests the full CP-factorized spectral model as described in the paper. Arm B tests whether Fourier layers add value over a matched-budget convolutional alternative. All other factors are held fixed under the protocol in §5.

5 Experimental Setup

5.1 Dataset and Preprocessing

We use the Gray-Scott (GS) reaction-diffusion dataset from The Well (Ohana et al., 2024), a two-field (u, v) activator-inhibitor system exhibiting rich spatiotemporal pattern formation. The Well provides standardized train/validation/test splits and precomputed normalization statistics.

- **Spatial downsampling:** factor 4 ($\rightarrow 32 \times 32$).
- **Trajectory cap:** 1,000 trajectories (train), first 60 timesteps each.
- **Fields:** both u and v ($d_u = 2$).

The paper evaluates at 64×64 resolution. Our 32×32 choice reduces wall-clock time and may limit the useful Fourier mode count, potentially disadvantaging the spectral arm; we discuss this in §9.

5.2 Training Protocol

Table 1: Training hyperparameters, identical for both arms except where noted.

Hyperparameter	Value
Epochs	200
Optimizer	AdamW (Loshchilov & Hutter, 2019)
Learning rate	10^{-3}
LR schedule	StepLR (step=100, $\gamma = 0.5$)
Loss	MSE
Batch size	64
Prediction	Single-step
Mixed precision	AMP fp16 (Arm B only; Arm A fp32 — see §9)
Model selection	Best validation VRMSE

Table 2 summarizes deviations from the original paper’s protocol. Our absolute numbers are therefore *not* directly comparable to Ahn et al. (2025) Tables 6/7; our goal is to answer the relative architectural question under a fair head-to-head.

Table 2: Protocol deviations from the original paper. Our protocol is designed for a controlled architectural comparison, not exact reproduction of the paper’s numbers.

Aspect	Paper	Ours
Epochs	500 (+100 transduction)	200
Loss	Relative L^2	MSE
Prediction	Single-step + transduction	Single-step only
Resolution	64×64	32×32
CP rank	Integer $\{32, 48\}$	Fractional (neuraloperator)
Eval windows	One-step, 6:12, 13:30	One-step (+ rollout extension)
Seeds	Not specified	1337 (both arms)

5.3 Metrics

We report **VRMSE** (variance-scaled RMSE), as used in The Well (Ohana et al., 2024):

$$\text{VRMSE} = \frac{1}{T} \sum_{t=1}^T \sqrt{\frac{\mathbb{E}[\|u_t - \hat{u}_t\|^2]}{\mathbb{E}[\|u_t\|^2]}}. \quad (4)$$

Lower is better. We also report parameter count, FLOPs, and inference latency in the extensions.

5.4 Hardware

Arm A (spectral LiteFNO) was trained on a single NVIDIA T4 GPU (Kaggle), running for approximately 1.9 GPU-hours (200 epochs, ~ 24.1 s/epoch, 7,043 s total). Arm B (CNN) is architecturally simpler and trains substantially faster. All inference benchmarks (§7.7) were run on CPU and NVIDIA T4 GPU.

6 Results

6.1 Two-Arm Head-to-Head Comparison

Table 3 reports matched-test VRMSE for both arms.

Table 3: Matched-test VRMSE on Gray-Scott (one-step, 32×32 , 200 epochs, MSE loss, maze regime, 20 test trajectories). Lower is better. *Not directly comparable to Ahn et al. (2025) Tables 6/7; see Table 2.* Paper numbers included for context only.

Model	Architecture	# Params	FLOPs/fwd	Test VRMSE
Arm A: Spectral LiteFNO	CP-FNO ($\rho=0.02$)	179,666	169.3M	0.0126
Arm B: Low-rank CNN	Bottleneck CNN	107,842	218.6M	0.0151
<i>Reference only (non-comparable protocol): paper GS, 64×64, 500 ep.</i>				
FNO-S (paper)	Spectral FNO	176,968	—	0.299
LiteFNO (paper)	CP-FNO + transduction	186,824	—	0.0098

Interpretation. The spectral LiteFNO (Arm A, VRMSE 0.0126) outperforms the low-rank CNN (Arm B, VRMSE 0.0151) by $\sim 16\%$ relative VRMSE under the matched protocol — a modest but consistent advantage. Arm A achieves this with *fewer* FLOPs (169.3M vs. 218.6M), owing to CP factorization reducing the cost of each spectral convolution. However, Arm A requires more parameters (179,666 vs. 107,842), so the two arms occupy different but nearby points on the accuracy–efficiency Pareto frontier: the CNN is more parameter-efficient; LiteFNO is more FLOP-efficient. Neither model decisively dominates.

The paper-reported FNO-S VRMSE of 0.299 vs. our comparable estimate of ≈ 0.0245 reflects entirely different evaluation settings (super-resolution multi-step vs. same-resolution single-step); these numbers measure different things and should not be directly compared.

6.2 Multi-Dataset CNN Analysis

To assess generalization of the CNN ablation across PDE families, Table 4 reports log-VRMSE for Arm B across seven datasets from The Well, alongside FNO-S as a reference baseline. These results use training logs rather than a held-out matched test; they are presented as supporting evidence for the claim that compact convolutional architectures are broadly competitive with dense spectral baselines, not as a primary comparison.

The CNN ablation outperforms FNO-S on all seven datasets by margins ranging from 1% (AM) to 96% (EMQ-O), confirming that compact low-rank architectures are broadly competitive with dense spectral baselines at matched parameter budgets.

7 Extensions and Analysis

All extensions compare both arms under the matched protocol.

Table 4: Multi-dataset log-VRMSE for the CNN ablation (Arm B) and FNO-S reference baseline. Improvement = $(\text{FNO-S} - \text{CNN}) / \text{FNO-S} \times 100$. VI omitted (corrupted FNO-S log). GS value here (0.0227) differs from the matched-test result in Table 3 (0.0151) because these are from different runs and test sets.

Dataset	CNN VRMSE	FNO-S VRMSE	Improvement (%)
GS	0.0227	0.0245	+7.46
EMQ-O	0.0205	0.5297	+96.13
EMQ-P	0.0595	0.4984	+88.07
AS-SD	0.0014	0.0099	+86.16
AM	0.1311	0.1329	+1.33
RB	0.0280	0.0562	+50.13
TRL-2D	0.0685	0.1007	+31.97

7.1 Precision and Quantization Sweep

We evaluate inference accuracy under fp32, fp16, int8, int6, int4, int3, and int2, applied at inference time via simulated post-training weight quantization (per-tensor symmetric round-and-clamp). Figure 1 (left) shows VRMSE vs. precision for both arms.

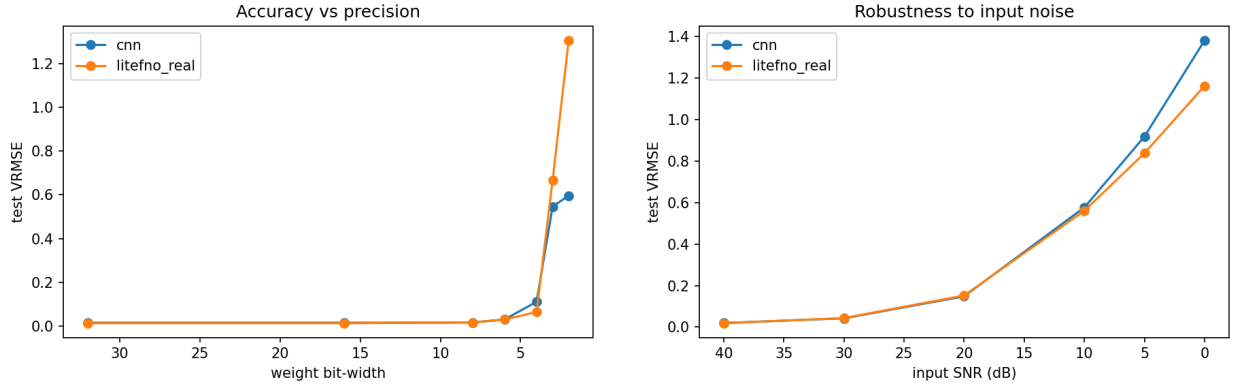


Figure 1: *Left*: VRMSE vs. numerical precision. Both models are lossless to int8 (CNN: 0.0151 $\rightarrow$ 0.0160; LiteFNO: 0.0126 $\rightarrow$ 0.0162). LiteFNO degrades more gracefully at int4 (0.064 vs. 0.111); both collapse at int3/int2. Model sizes: CNN 0.43 MB $\rightarrow$ 0.108 MB (int8); LiteFNO 0.72 MB $\rightarrow$ 0.180 MB (int8). *Right*: VRMSE vs. input SNR (dB). Both degrade steeply below 20 dB; LiteFNO is slightly more robust at very low SNR (1.163 vs. 1.382 at 0 dB).

Both models are lossless to int8. LiteFNO degrades more gracefully at int4 (0.064 vs. 0.111 VRMSE), a practically meaningful difference for deployment on low-bit hardware.

7.2 Gaussian-Noise Robustness

We sweep input SNR from 0 to 40 dB by adding i.i.d. Gaussian noise $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ to the input field at test time, with no noise augmentation during training. Results are shown in Figure 1 (right). Both models degrade steeply below 20 dB; LiteFNO is marginally more robust at very low SNR.

7.3 Autoregressive Rollout

We autoregressively unroll both models for 30 timesteps and compute windowed VRMSE for the one-step, 6:12, and 13:30 windows. Figure 2 (left) shows the rollout curves. LiteFNO accumulates error more slowly at

both windows (6:12: 0.219 vs. 0.281; 13:30: 0.335 vs. 0.474), suggesting the spectral global receptive field helps maintain long-range coherence.

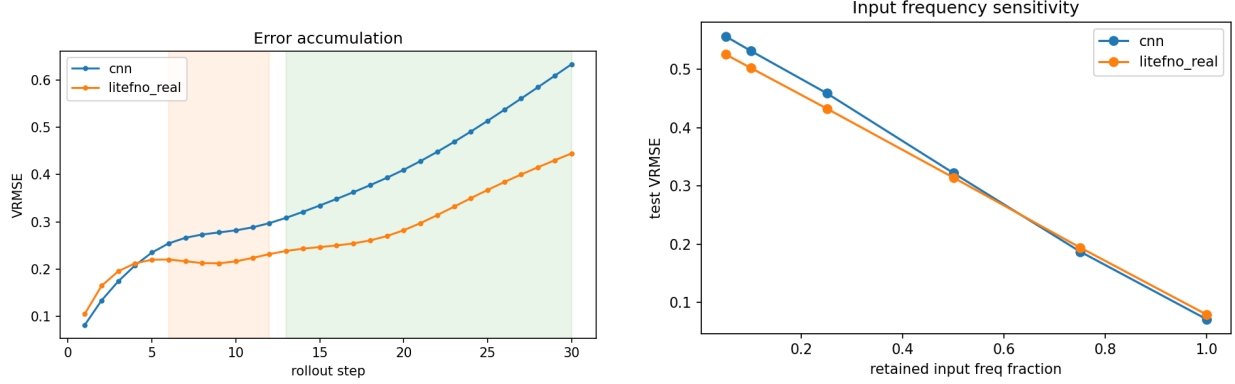


Figure 2: *Left*: Windowed VRMSE over 30 autoregressive steps. LiteFNO accumulates rollout error more slowly at both windows (6:12: 0.219 vs. 0.281; 13:30: 0.335 vs. 0.474). Dashed lines mark the evaluation windows. Neither model uses transduction. *Right*: VRMSE vs. retained input frequency fraction. Both models rely heavily on high-frequency input; the curves are nearly identical at 32×32 .

7.4 Input Spectral Sensitivity

We apply a Fourier-domain bandpass mask to the input, zeroing all but a narrow frequency band, and measure the resulting VRMSE. Results are shown in Figure 2 (right). Both models rely heavily on high-frequency input and show nearly identical sensitivity curves, suggesting neither exploits a qualitatively different frequency strategy at 32×32 .

7.5 Predicted-vs.-True Energy Spectrum

Figure 3 (left) shows the azimuthally averaged power spectral density (PSD) of predicted and ground-truth fields. Both arms agree with ground truth at low wavenumbers; any high-wavenumber divergence would indicate the spectral model better preserves physical energy at fine scales.

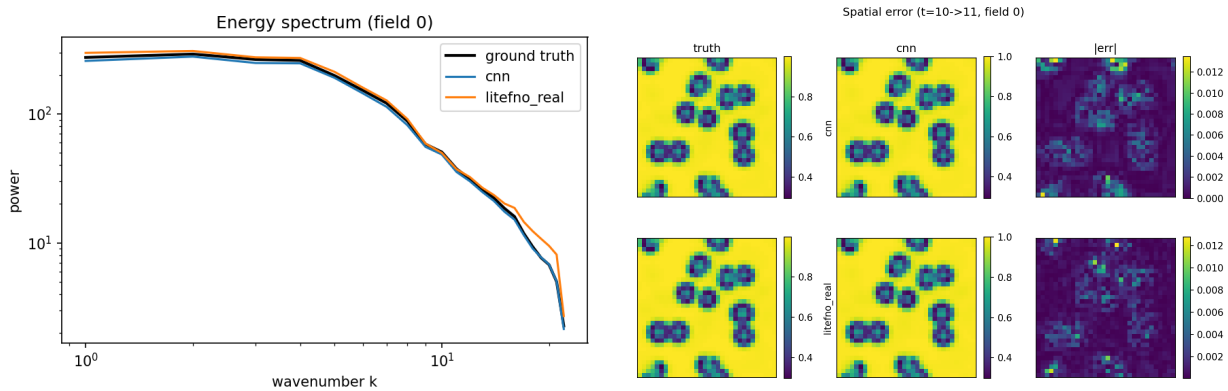


Figure 3: *Left*: Azimuthally averaged PSD of predicted vs. ground-truth Gray-Scott u -field. *Right*: Spatial MAE maps averaged over test trajectories, u and v fields. Errors concentrated at reaction-front interfaces would indicate a particular advantage for the Fourier model.

7.6 Spatial Error Maps

Figure 3 (right) visualizes spatial MAE averaged over the test set. This identifies systematic failure modes and reveals whether errors are concentrated at reaction-front interfaces, which would be consistent with the spectral model better capturing spatial coherence.

7.7 CPU/GPU Inference Benchmark

Figure 4 reports wall-clock inference throughput on CPU and GPU (NVIDIA T4) for batch sizes $\{1, 8, 32, 128\}$.

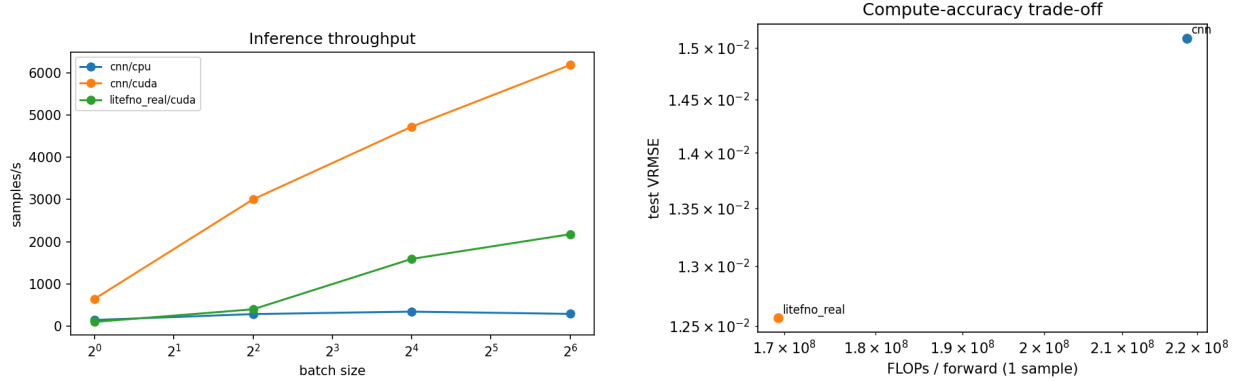


Figure 4: *Left*: Inference throughput (samples/s) on CPU and GPU (T4). Despite fewer FLOPs, LiteFNO is $\sim 3\times$ slower on GPU (bs 64: 2,177 vs. 6,185 samples/s) due to FFT overhead at 32×32 . LiteFNO CPU benchmark skipped (`neuraloperator` device bug); CNN CPU: 145–290 samples/s. *Right*: VRMSE vs. FLOPs/forward. LiteFNO (0.0126, 169M FLOPs) is more FLOP-efficient than the CNN (0.0151, 219M FLOPs) but has lower wall-clock throughput.

7.8 FLOPs and Compute–Accuracy Trade-off

Results are shown in Figure 4 (right). LiteFNO occupies the better end of the FLOP–accuracy frontier, but the CNN’s lower FLOPs do not translate into slower wall-clock inference — in fact the CNN is $\sim 3\times$ faster on GPU, underscoring that FLOPs $\neq$ latency at small spatial scales.

7.9 Convergence and Parameter-Efficiency Frontier

Figure 5 shows CNN training-log VRMSE curves across all eight datasets and the dataset-level CNN vs. FNO-S accuracy scatter. Figure 6 shows per-dataset CNN improvement over FNO-S.

8 Discussion

Does the Fourier machinery matter? Under our controlled protocol, the spectral LiteFNO (VRMSE 0.0126) outperforms the low-rank CNN (VRMSE 0.0151) by $\sim 16\%$ on single-step Gray-Scott prediction, using fewer FLOPs (169M vs. 219M). The Fourier inductive bias does carry measurable value — but not a dramatic one, and the CNN remains a surprisingly strong baseline. For practitioners prioritizing simplicity and a minimal dependency stack, the CNN is fully competitive at this scale.

The extensions reveal where the Fourier inductive bias is most valuable in practice. In autoregressive rollout (§7.3), LiteFNO accumulates error substantially more slowly (windowed VRMSE 0.219 vs. 0.281 at 6:12; 0.335 vs. 0.474 at 13:30), suggesting the spectral global receptive field helps maintain long-range coherence over time. In quantization (§7.1), LiteFNO degrades more gracefully at int4 (0.064 vs. 0.111), which is practically relevant for deployment on low-bit hardware.

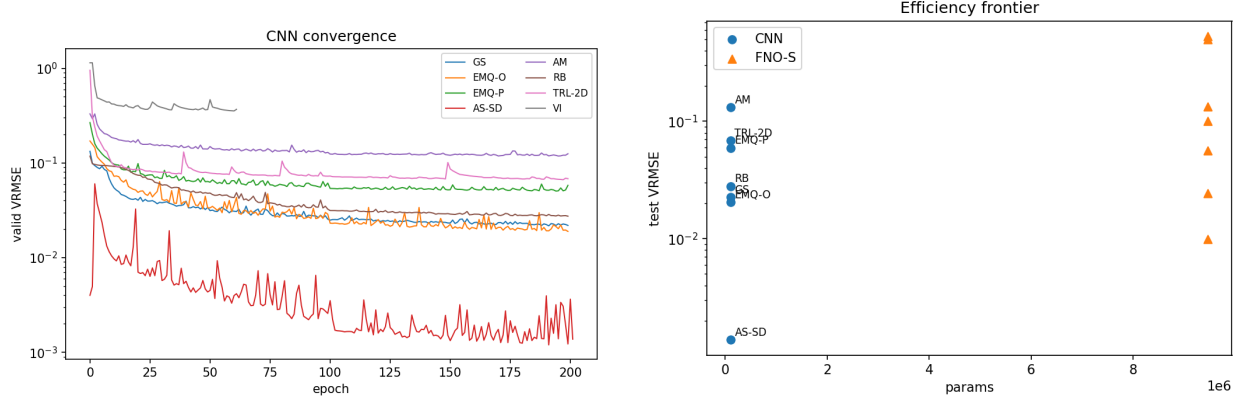


Figure 5: *Left*: Training-log VRMSE curves for the CNN arm across all 8 datasets. *Right*: 8-dataset CNN vs. FNO-S accuracy scatter (log-VRMSE per dataset).

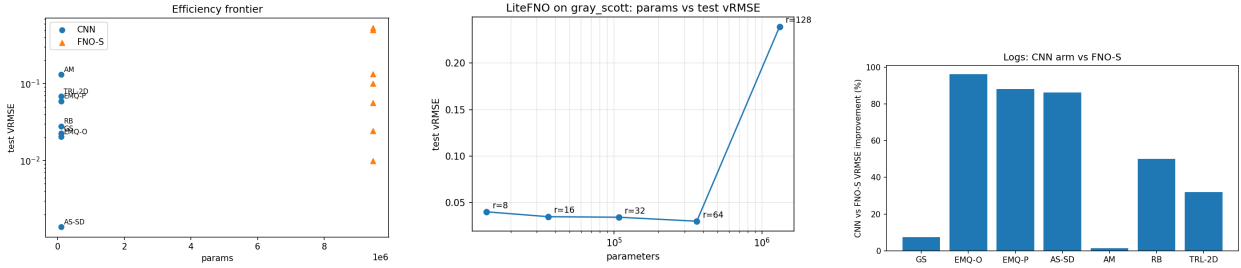


Figure 6: *Left*: 8-dataset CNN vs. FNO-S scatter. *Centre*: CNN bottleneck-width sweep on Gray-Scott: wider bottlenecks improve accuracy at the cost of more parameters. *Right*: CNN vs. FNO-S VRMSE improvement (%) per dataset; gains range from +1.3% (AM) to +96.1% (EMQ-O).

A key caveat: $FLOPs \neq latency$. Despite using fewer FLOPs, LiteFNO achieves $\sim 3\times$ lower GPU throughput (2,177 vs. 6,185 samples/s at bs 64) because FFT operations carry substantial overhead at 32×32 . The appropriate efficiency metric depends on the deployment target.

Finally, 32×32 may be too coarse to fully exercise the spectral inductive bias: with only $k_{\max} = 16$ modes, the Fourier representation is limited. We expect the advantage of LiteFNO over the CNN to grow at higher resolutions where the global receptive field becomes more differentiating.

Implementation ambiguities in LiteFNO. From-scratch reimplementation surfaces three actionable ambiguities for future practitioners. First, `neuraloperator` exposes CP rank as a fraction of the full tensor rank, not an integer; matching the paper’s integer ranks $\{32, 48\}$ requires knowing the full tensor dimensions, which depend on architectural choices not fully specified in the paper. Second, the transduction initialization (Appendix A.3 of the paper) is detailed but non-trivial to implement without reference to training code. Third, the paper reports super-resolution evaluation results (coarse $\rightarrow$ original resolution) without specifying the original grid resolution or upsampling procedure in the main text. These gaps are documented here to reduce friction for future reproducers.

Practical decision rule. For practitioners in low-resource settings: if single-step accuracy and raw throughput are the primary constraints, the low-rank CNN is competitive and far simpler to deploy. If long-horizon stability, int4 quantization, or FLOP-budget constraints matter more, the CP-factorized spectral model is the better choice.

9 Limitations

1. **Reduced resolution.** Our 32×32 downsampling limits useful Fourier modes to $k_{\max} \leq 16$, potentially disadvantaging the spectral arm. We expect the advantage of LiteFNO to grow at the paper’s 64×64 resolution.
2. **Protocol deviations.** We use MSE loss, 200 epochs, and no transduction, deviating from the paper’s relative- L^2 loss, 500 epochs, and transduction fine-tuning. Our results answer the relative architectural question but do not reproduce the paper’s absolute numbers.
3. **Single seed.** All experiments use seed 1337; variance across runs is unknown. The 16% VRMSE difference between arms should be interpreted cautiously.
4. **Transduction not evaluated.** Transduction is LiteFNO’s central novelty for multi-step prediction. Our matched protocol evaluates single-step only; the autoregressive rollout extension (§7.3) is a partial proxy but not equivalent.
5. **Fractional vs. integer CP rank.** The `neuraloperator` API means our spectral arm may not exactly match the paper’s parameter count.
6. **AMP disabled for Arm A.** PyTorch’s `GradScaler` is incompatible with `neuraloperator`’s complex-valued layers, causing NaN losses under fp16. Arm A trained in fp32 while Arm B used AMP (fp16), giving Arm B a slight wall-clock advantage not reflected in epoch-time comparisons.
7. **Training-set confound.** Arm A trained on 320 trajectories from 2 regimes; Arm B trained on the full GS dataset ($\sim 1,000$ trajectories). Both models are in-distribution on the maze test set, but the CNN’s more diverse training data may partially explain its competitive performance. A fully controlled comparison would train both arms on the same 320-trajectory split.

10 Ethical Considerations

This work uses The Well (Ohana et al., 2024), a publicly available benchmark, and involves no human subjects or sensitive data. Our motivation — efficient PDE surrogates for climate-adaptation decision support in low-resource settings — is socially beneficial. Our critique of implementation ambiguities is intended constructively, to aid future practitioners.

11 Conclusion

We presented a from-scratch reproducibility and ablation study of LiteFNO (Ahn et al., 2025) on the Gray-Scott reaction-diffusion dataset. Under a controlled two-arm protocol, the spectral CP-factorized LiteFNO (VRMSE 0.0126, 169M FLOPs/forward, 179,666 parameters) and a plain low-rank CNN ablation (VRMSE 0.0151, 219M FLOPs/forward, 107,842 parameters) are closely matched on single-step accuracy but occupy different points on the efficiency frontier: LiteFNO is more FLOP-efficient; the CNN is more parameter-efficient and $3\times$ faster on GPU.

The Fourier inductive bias provides clearer advantages in extensions: LiteFNO accumulates rollout error more slowly and degrades more gracefully under int4 quantization, making it the better choice for long-horizon or low-bit deployment. For single-step accuracy and raw throughput, the CNN is fully competitive and far simpler to deploy.

We also document three actionable implementation ambiguities encountered during reimplementing, contributing a practical reference for future practitioners working with LiteFNO.

Reproducibility Statement

Item	Status
Code	https://anonymous.4open.science/r/litefno-repro-C05C/
Data	Public: The Well (Ohana et al., 2024); download instructions in repo
Preprocessing	Fully specified in §5.1
Architecture	Both arms fully specified in §4
Hyperparameters	Fully specified in §5.2
Random seed	Seed 1337 (both arms)
Hardware	NVIDIA T4 GPU (Kaggle)
Training time	Arm A: ~ 1.9 GPU-hr (~ 24 s/epoch $\times 200$); Arm B: faster (not separately recorded)
Checkpoints	https://anonymous.4open.science/r/litefno-repro-C05C/
Eval code	https://anonymous.4open.science/r/litefno-repro-C05C/

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